

SIMULATION OF AGENT BASED OF TURKEY'S EMISSION TRADING SYSTEM

By

Oğuz Gökdemir

Melis Mağden

İbrahim Hakkı Keleş

Department Of Industrial Engineering

Advisor

Asst. Prof. Dr. Deniz Efendioğlu



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WHY CARBON PRICING?

1. INTERNATIONAL CLIMATE COMMITMENTS

- Paris Agreement (2015)
- Turkey ratification (2021)
- Net-zero target (2053)

2. ECONOMIC EFFICIENCY OF MARKET BASED INSTRUMENTS

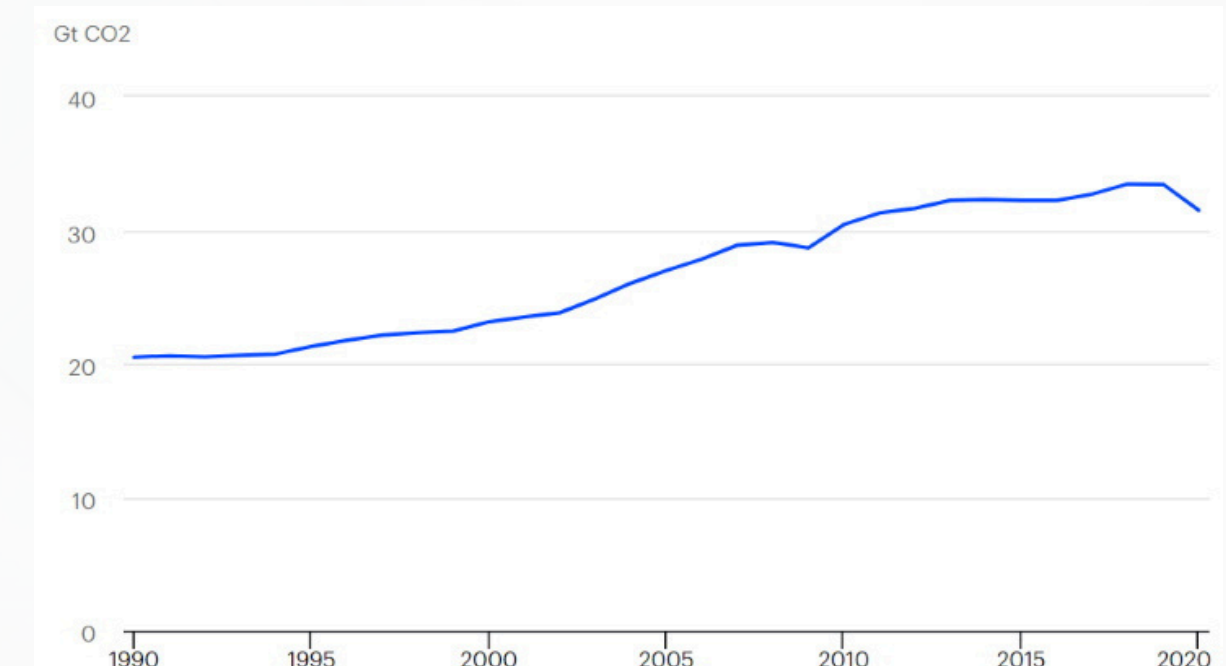
- ETS: Cost effective & flexible
- Cap and trade principle
- Incentivizes efficient reduction

3. EU ETS SUCCESS

- Operational since 2005
- Proven effectiveness
- 35% emission reduction in covered sectors

4. GLOBAL DIFFUSION & INTERNATIONAL CARBON MARKETS

- China, South Korea, California
- International carbon markets
- Turkey's integration imperative



TURKEY'S CLIMATE CHALLENGE

**EMISSION
GROWTH**
+144%
(1990-2020)
558.3 Mt CO₂e

**ENERGY
DOMINANCE**
72%
**Share of total
emission**

EU CBAM PRESSURE
2026
Full implementation
Export risk

NET-ZERO TARGET
2053
National commitment

WHAT IS AN EMISSION TRADING SYSTEM?

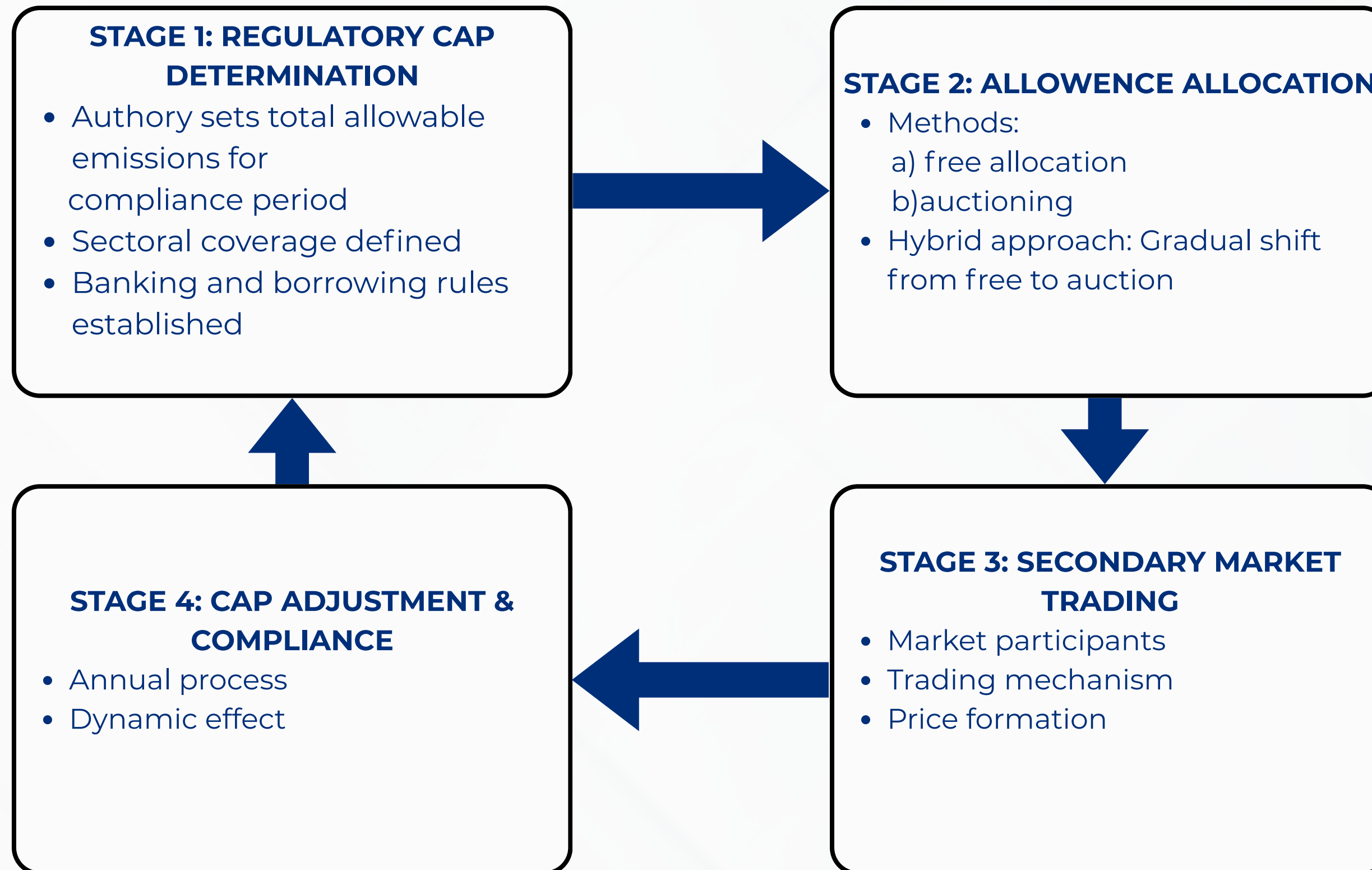
- A market-based policy instrument designed to reduce greenhouse gas emissions in a cost-effective and flexible manner through a cap-and-trade mechanism.

CAP SETTING

**ALLOWENCE
TRADING**

**CAP
TRAJECTORY**

HOW ETS WORKS ?



TURKEY'S THREE PROBLEMS

- **PROBLEM 1 : INSTITUTIONAL GAP**

Absence of Domestic Carbon Pricing

- **PROBLEM 2 : CBAM EXPOSURE**

Competitiveness Risk under EU Regulation

- **PROBLEM 3 : FISCAL LEAKAGE**

Revenue Outflow & Policy Sovereignty

TURKEY'S POLICY LANDSCAPE

No mandatory ETS

- Voluntary mechanisms only
- Preparatory phase

MRV regulation (2015)

- Not integrated with trading
- Selected facilities

Administrative approach

- Command-and-control
- No carbon price signal

SECTORAL EMISSION STRUCTURE

ENERGY SECTOR: 72% of total

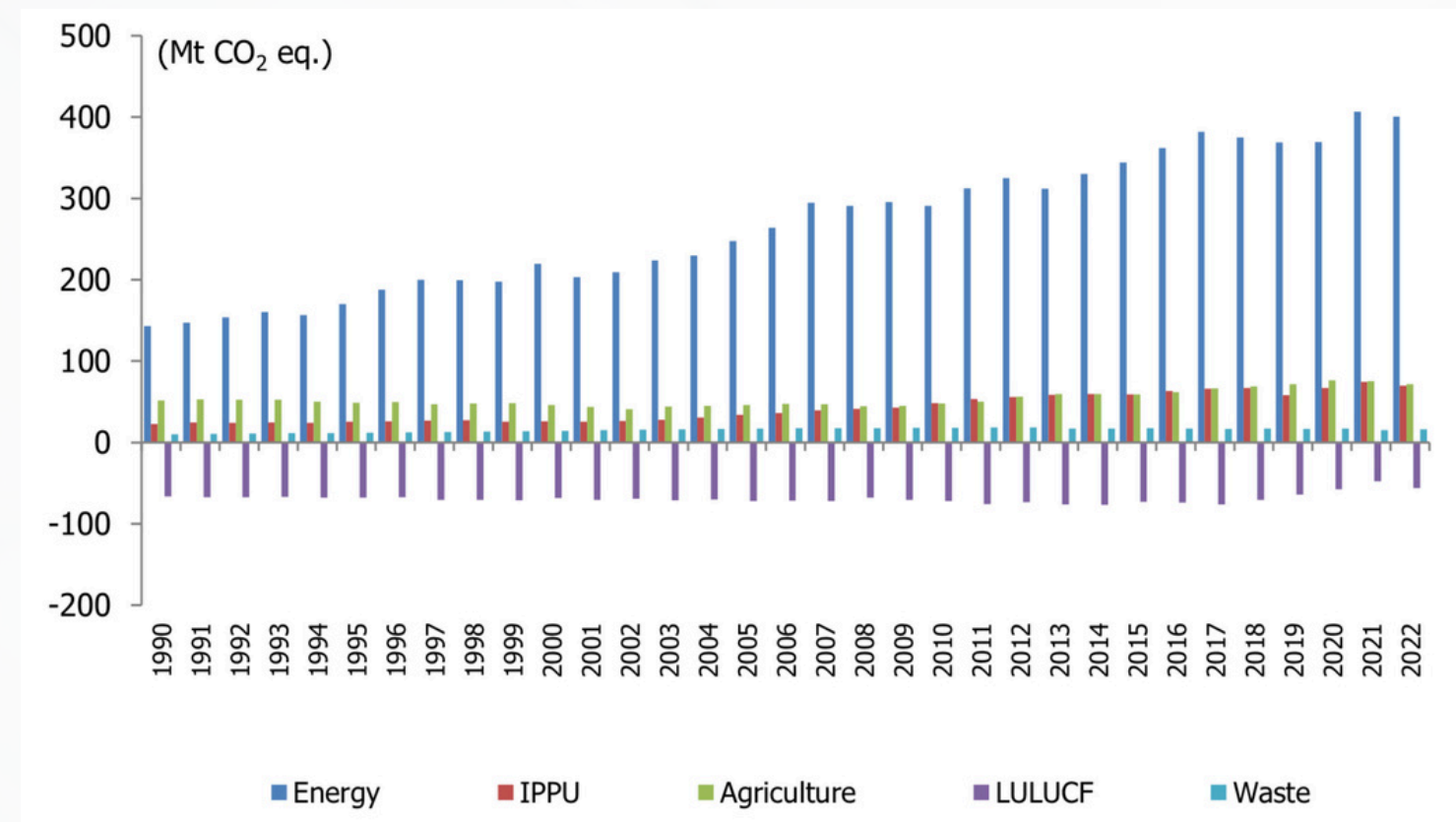
- Electricity & heat: 35.9%
- Transport: 15.8%
- Manufacturing: 13.7%
- Other energy: 6.6%

OTHER SECTORS:

- Industrial processes: 12.8%
- Agriculture: 10.7%
- Waste: 4.5%

REGIONAL:

- Marmara Region: ~40% industrial output



EU CBAM IMPLICATIONS

2023-2025 → Transition

2026 → Full (payment)

SECTORS: Cement, Steel, Aluminum, Fertilizers, Electricity, Hydrogen

WITHOUT DOMESTIC ETS:

- €4Bn/year cost estimate
- Paid at EU border
- Revenue to EU
- Zero control

WITH DOMESTIC ETS:

- Pay difference only
- Revenue stays domestic
- Policy sovereignty
- +€2.6Bn/year benefit

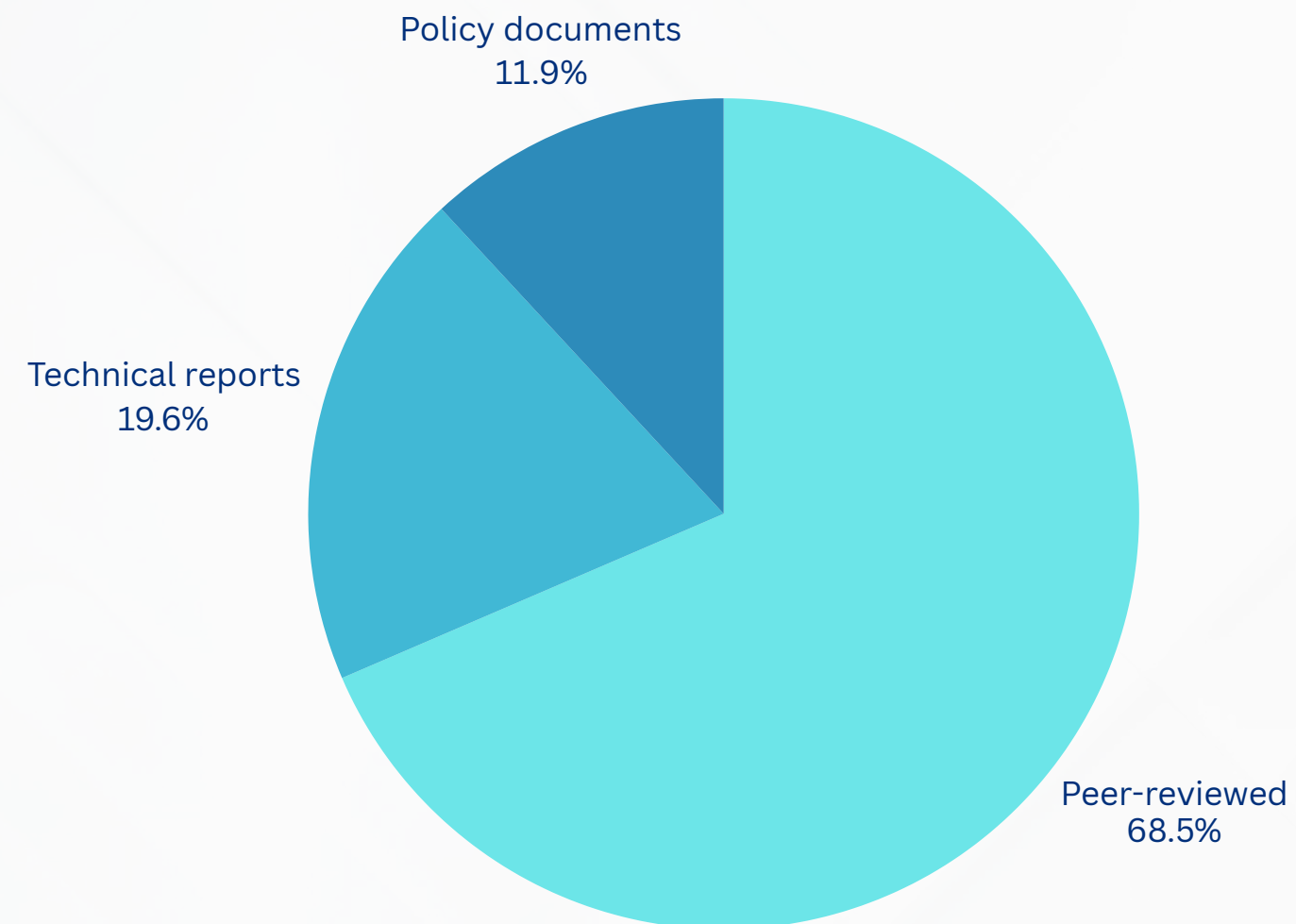
LITERATURE REVIEW

Total sources researched: 133

Themes:

- Modeling approaches
- Carbon pricing & ETS
- Agent-based modeling

Period: 1990-2024



MODELING COMPARISON

Approach
Top-Down(CGE)
Bottom-up (TIMES)
Machine Learning (ANN)
Agent-based (ABM)

WHY AGENT-BASED MODELING?

01

HETEROGENEITY

- Firms differ fundamentally
- Abatement costs vary
- Technology portfolios differ

02

DYNAMICS

- Temporal evolution (2026-2035)
- Adaptive mechanisms

03

INTERACTIONS

- Allowance market dynamics
- Strategic behaviors
- Market feedback loops

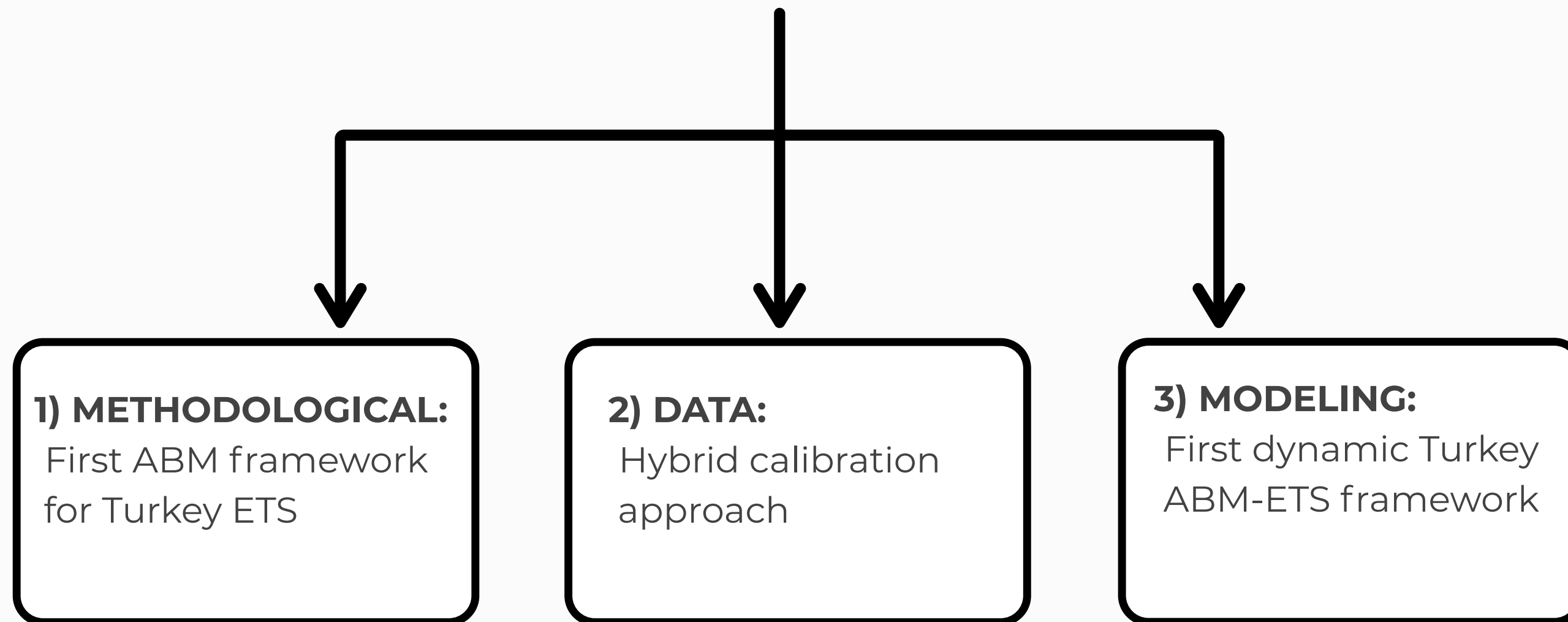
04

EMERGENCE

- Carbon price
- Total emission reductions
- Sectoral patterns
- Regional clusters

RESEARCH GAP IDENTIFICATION

OUR CONTRIBUTION ADDRESSES ALL THREE GAPS:



RESEARCH METHODOLOGY FRAMEWORK

EPISTEMOLOGY

- Positivist stance
- Quantitative methods
- Cause-effect relationships

PRIMARY METHOD

- Agent-Based Modeling (ABM)
- Simulation modeling approach
- Scenario comparison
- Policy experimentation

SECONDARY METHODS

- Literature review
- Policy document analysis
- Statistical evaluation (planned)

DATA SOURCES

- TurkStat
- EPDK
- TEİAŞ
- Literature

SIMULATION DESIGN

- Period: 2026-2035 (10 years, annual)
- Scenarios: 5 policy configurations
- Agents: Facilities, Market Operator, Households

THREE AGENT TYPES

INDUSTRIAL FACILITIES

- Will decide whether to reduce emissions or purchase allowances based on cost considerations

HOUSEHOLDS

- Will represent changes in energy consumption patterns and indirect effects.



MARKET OPERATOR

- Will manage the allowance system and update prices

FIVE POLICY SCENARIOS (2026-2035)

Scenario	Allocation
1. BAU	N/A Baseline
2. MODERATE ETS	80% free 20% auction
3. STRICT ETS	50% free 50% auction
4. CARBON TAX	€50/ton (flat)
5. ETS + INVESTMENT	80% free + subsidy

SIMULATION PARAMETERS:

- Period: 2026-2035 (10 years)
- Time step: Annual
- Total runs: 50 (5 scenarios × 10 years)

COMPARISON METRICS (PLANNED):

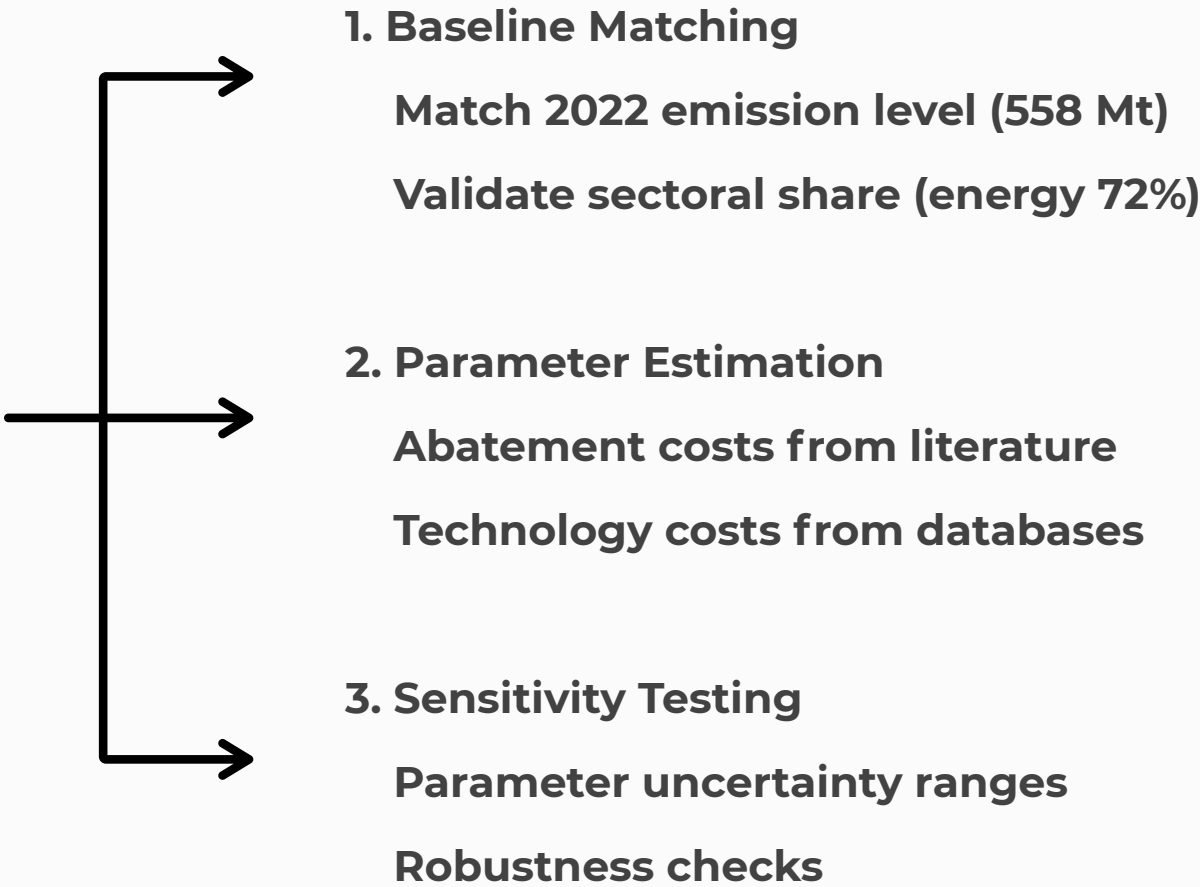
- Emission levels (Mt CO₂)
- Carbon prices (€/ton)
- Economic costs (% GDP)
- Revenue generation (€ billion)

DATA SOURCES & CALIBRATION APPROACH

DATA INTEGRATION:

Data Type	Source	Coverage	Model Use
National GHG	TurkStat	1990-2022	Baseline calibration
Energy Sector	EPDK	2010-2024 monthly	Sectoral breakdown
Electricity	TEİAŞ	2010-2024 hourly	Power sector
Facility MRV	UNFCCC NIR	2019-2023 selected	Large emitters
Abatement cost	Literature	sectoral estimates	Cost curves

CALIBRATION STEPS (PLANNED):



EXPECTED ANALYTICAL OUTCOMES

ENVIRONMENTAL EFFECTIVENESS

Emission Trajectory Projections:

- Business-as-usual (BAU): Continued growth trajectory
Projected: ~650 Mt CO₂e by 2035
- Moderate ETS scenario: Approximately 30% reduction
Cap stringency: -2% annual reduction
- Strict ETS scenario: Approximately 35-40% reduction vs. BAU
Cap stringency: -3% annual reduction
Target alignment: Turkey's NDC commitment (41% by 2030)

Evaluation criteria:

- Consistency with Paris Agreement obligations
- Contribution to 2053 net-zero pathway
- Sectoral distribution of emission reductions

ECONOMIC EFFICIENCY

Carbon Price Dynamics:

- Moderate ETS
- Strict ETS
- Carbon tax (alternative)

Revenue Generation Potential:

- Estimated auction proceeds: €2-4 billion annually
- Allocation priorities:
 - Green technology R&D and deployment
 - Renewable energy infrastructure subsidies
 - Just Transition support (coal-dependent regions)
 - Energy efficiency program financing

GDP Impact Assessment:

- Moderate ETS: -0.5% to -1.0% GDP
- Strict ETS: -1.0% to -2.0% GDP
- Evaluation: Within manageable macroeconomic threshold
- Comparison: EU ETS experience (similar range)

EXPECTED POLICY INSIGHTS

1. ETS FEASIBILITY

- 35-40% emission reduction achievable
- Compatible with NDC and net-zero pathway

2. GRADUAL DESIGN IMPERATIVE

- Start moderate → Tighten progressively
- Minimize economic disruption

3. REVENUE RECYCLING ESSENTIAL

- €2-4 billion for green transition
- Social acceptance and equity

4. CBAM COMPATIBILITY ADVANTAGE

- Net benefit: +€2.6 billion/year
- Policy sovereignty retained

THREE-PHASE IMPLEMENTATION:

- Phase 1 (2026-27): Energy, 80% free
- Phase 2 (2028-30): Add industry, 50% free
- Phase 3 (2031-35): All sectors, 70% auction

RESEARCH CONTRIBUTIONS

1. THEORETICAL

- First ABM application to Turkey ETS
- Extended agent-based climate policy modeling

2. EMPIRICAL

- First Turkey-specific ETS simulation
- Integrated fragmented national data sources

3. METHODOLOGICAL

- Hybrid calibration for data constraints
- Template for emerging ETS countries

4. POLICY

- Decision-support tool for policymakers
- Three-phase implementation roadmap
- Quantified benefits

RESEARCH SUMMARY

- Our research develops Turkey's first agent-based ETS simulation framework to provide evidence-based guidance for climate policy design before the 2026 EU CBAM deadline. We systematically reviewed 133 studies, identified a critical research gap, and designed a modeling framework with three agent types and five policy scenarios. Our findings demonstrate that Turkey can transform the CBAM threat into an economic opportunity while achieving significant emission reductions aligned with NDC targets. In the first term, we completed the conceptual design, data integration and initial model setup; we are currently in the development and testing phase, with simulation runs and comprehensive analysis planned for completion in the second term to finalize policy recommendations guiding Turkey's three phase ETS establishment.

THANKS FOR ATTENTION

Friday

January 16, 2026

GRADUATION PROJECT PRESENTATION

3:15 PM

By

Oğuz Gökdemir

Melis Mağden

İbrahim Hakkı Keleş

Advisor

Asst.Prof.Dr.Deniz Efendioğlu